INCIDENT RESPONSE REPORT — Task 2 (SIEM Analysis Using Splunk)

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1. Overview

This report documents the detection and analysis of suspicious authentication activity using Splunk SIEM.
The purpose of this task was to ingest log data, perform alert triage, identify malicious behavior, classify the incident, and recommend mitigation steps.

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2. Log Source Details

- **Log File Used:** `windows_security_logs.txt`
- **Type of Logs:** Windows Security Authentication Events
- **Index Created:** `task2_index`
- **Sourcetype:** `log2metrics_keyvalue`

 Screenshot 1 — Log Upload in Splunk
➡ (See: `/screenshots/upload.png`)

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3. Objective of the Task

- Monitor security alerts using SIEM
- Analyze authentication events
- Identify suspicious login activity
- Detect brute-force login patterns
- Classify the incident (severity, type, attacker behavior)
- Create dashboards/visualizations
- Draft an SOC-style incident report

4. Searches Performed (Splunk Queries)

✔ 4.1 View All Events

```
index="task2_index"
```

✔ 4.2 Failed Login Attempts

```
index="task2_index" "Failed Login"
```

✔ 4.3 Successful Login Attempt

```
index="task2_index" "Successful Login"
```

✔ 4.4 Total Failed Login Count

```
index="task2_index" "Failed Login"
| stats count
```

✔ 4.5 Failed Logins by IP Address

```
index="task2_index" "Failed Login"
| stats count by IpAddress
```

✔ 4.6 EventID Breakdown (4625 vs 4624)

```
index="task2_index"
| stats count by EventID
```

 Screenshot 2 — Query Results
➡ (See: `/screenshots/query_results.png`)

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5. Visualizations

✔ 5.1 Failed Login Attempts by IP (Bar Chart)

```
index="task2_index" "Failed Login"
| stats count by IpAddress
```

✔ 5.2 EventID Distribution (Pie Chart: 4625 vs 4624)

```
index="task2_index"
| stats count by EventID
```

 Screenshot 3 — Visualizations
➡ (See: `/screenshots/visualizations.png`)

6. Findings (Alert Analysis)

✔ Event Summary

EventID	Meaning	Count
4625	Failed Login	**3**
4624	Successful Login	**1**

✔ Identified Behavior

- Same account targeted: **admin**
- Same external IP used: **185.34.55.1**
- Multiple failed attempts within seconds
- Followed by a successful login

□ **This is a classic brute-force attack pattern.**

□ *Screenshot 4 — Failed + Successful Events*
➡□ (See: `/screenshots/event_timeline1.png and event_timeline2.png`)

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7. Incident Classification

Incident Type:

Brute-Force Authentication Attack

MITRE ATT&CK Mapping:

T1110: Brute Force

Severity Level:

High

Reason for Severity:

- Admin account targeted
- Password successfully compromised
- Multiple rapid failed → successful login
- Same external attacker IP

This indicates **account compromise**.

8. Recommended Remediation Actions

✔ Immediate Actions

- Block attacker IP: **185.34.55.1**
- Force password reset of “admin”
- Review systems accessed after compromise

✔ Policy-Level Fixes

- Enable **Account Lockout Policy**
- Enable **Multi-Factor Authentication (MFA)**
- Restrict admin login from external IPs
- Increase logging & monitoring

✔ Monitoring Recommendations

- Alert SOC on 3+ failed logins within 1 minute
- Continuous monitoring for EventID 4625 surges

□ 9. Screenshots

All screenshots stored in:

```
/screenshots/
```

Include these:

- upload.png
- search_all_events.png
- failed_logins.png
- successful_login.png
- eventid_breakdown.png
- bar_chart_failed_logins.png
- pie_chart_successful_login.png
- pie_chart_account_name.png
- pie_chart_Status.png
- event_timeline1.png
- event_timeline2.png

□ 10. Conclusion

A brute-force attack originating from **IP 185.34.55.1** was detected.
The attacker attempted multiple incorrect passwords, eventually gaining successful access to the “admin” account.
This incident is classified as **High Severity**, requiring immediate response actions to prevent system compromise.

Splunk SIEM effectively helped identify, analyze, and visualize the attack pattern.